

Eniac's Discount Strategy

Welcome to your second data project! You will keep working for Eniac —the e-commerce tech company— as a Data Analyst. This time you will work with internal data, which is not anonymised... but a bit more chaotic! The complexity of these tasks will require you to use Python, rather than SQL.

The company has high hopes put into the possibilities that come with Data Analysis, and they are especially hopeful that your work can finally settle an ongoing debate: whether or not it's beneficial to discount products.

- The Marketing Team Lead is convinced that offering discounts is beneficial in the long run. She believes discounts improve customer acquisition, satisfaction and retention, and allow the company to grow.
- The main investors in the Board are worried about offering aggressive discounts. They have pointed out how the company's recent quarterly results showed an increase in orders placed, but a decrease in the total revenue. They prefer that the company positions itself in the quality segment, rather than competing to offer the lowest prices in the market.

Business questions

Your analytical and business skills are needed to provide clarity in the following aspects:

- How many products are being discounted?
- How big are the offered discounts as a percentage of the product prices?
- How do seasonality and special dates (Christmas, Black Friday) affect sales?
- How should products be classified into different categories to simplify reports and analysis?
- What is the distribution of product prices across different categories?
- How could data collection be improved?

At the end of this project (your instructor will set the exact date), you are expected to give a presentation to the company board and present the results of your analysis. The presentation needs to be concise, but also engaging and convincing.

A problem with the data

Before you get too deep into the business questions, there is some bad news we need to share with you: the data appears to have some inconsistencies —it must have been corrupted.

The Database Administrator, Khader, blames the buggy pipeline that connects the online store with the database, while the Software Engineer, Lina, claims the issues have to do with wrong encodings and bad maintenance practices in our database. As Khader and Lina argue with one another and try to fix the data source, the company board does not have time to lose: they want your business insights.

In short, you will have to assess the data quality of the dataset and take on many data cleaning tasks before you can even start to think about the business questions.

How mad should you be about that? A popular figure shared all the time in Data communities is that Data Analysts and Scientists spend 80% of their time cleaning and preparing data. Ironically, there is no reliable data source for that number, but if you just talk with people in the field they will all tell you it's a major part of the job. There is little merit in applying advanced methods and fancy solutions to corrupted and unreliable datasets. The real value good Data Analysts and Scientists can bring is usually to make dirty data usable and trustable — with the necessary disclaimers.

Skills

Transform a messy dataset into a clean & trustable source ready for insight-extraction.

Data Cleaning & Quality

Using Python, create professional looking plots that convey a clear message

Data Visualization

Craft a narrative that makes your work memorable and drives business action

Data Storytelling

Tools



Environment for Notebooks



Programming Language



Modules

Deliverable

A 5-minute group **presentation**, created as the Eniac Data Analytics Team, on the topic of **product discounts** at Eniac.